Breast Cancer Detection Using Classical Machine Learning: A Statistically Grounded Comparison of Decision Trees, Ensembles and Baseline Classifiers

Udit Asthana
Dept. of Computer Science and Engineering
Manipal Institute of Technology, MAHE
Roll No. 31
Reg. No. 240905310

C. Nagesh Prabhu
Dept. of Computer Science and Engineering
Manipal Institute of Technology, MAHE
Roll No. 2
Reg. No. 240905014

Mahad Sheikh
Dept. of Computer Science and Engineering
Manipal Institute of Technology, MAHE
Roll No. 36
Reg. No. 240905382

Abstract* - Early and reliable separation of malignant from benign breast masses is central to timely treatment, and a missed malignancy is the costliest error a diagnostic aid can make. This paper compares eight classical classifiers, namely ID3, C4.5, CART, Random Forest, XGBoost, Logistic Regression, K-Nearest Neighbors and Support Vector Machine, on the Wisconsin Diagnostic Breast Cancer dataset of 569 fine-needle aspirate samples. ID3 and C4.5 were implemented from scratch. To make the comparison fair, every model was evaluated on the same 50 test folds of a ten-times repeated, stratified five-fold nested cross-validation, with hyperparameters tuned inside each fold by a recall-weighted criterion, and differences were tested with a variance-corrected paired t-test. Logistic Regression achieved the highest mean accuracy (0.974), recall (0.958) and Matthews correlation coefficient (0.945), missing about 9 of 212 malignant cases per pass over the data, but its lead over the Support Vector Machine and XGBoost was not statistically significant. All three single decision trees were significantly worse and missed about 21 malignant cases per pass. A conventional single train-test split gave the same ordering. The work provides a reproducible, statistically grounded benchmark, not a clinical tool.

Index Terms* - Breast cancer detection, classical machine learning, decision trees, nested cross-validation, statistical model comparison

INTRODUCTION

Breast cancer is the most frequently diagnosed cancer in women worldwide. Global estimates for 2020 counted about 2.3 million new cases and roughly 685,000 deaths [1], and the 2022 GLOBOCAN update shows that the disease remains a leading contributor to cancer incidence and mortality [2]. Outcomes depend heavily on how early a malignant mass is recognised, so accurate discrimination between benign and malignant masses is a clinical priority.

Fine-needle aspiration (FNA) is a minimally invasive procedure whose digitised images can be analysed to compute quantitative descriptors of cell nuclei, such as radius, texture, perimeter, area and concavity. The multisurface pattern-separation work of Wolberg and Mangasarian [3] showed that such cytological measurements can separate benign from malignant specimens, and the Wisconsin data that grew out of this work have become a standard testbed for classifiers.

Recent surveys show that computer-aided breast cancer diagnosis spans deep learning systems for histopathology and genomic data [4] as well as machine learning methods more broadly [5]. Deep models typically need large labelled image collections and are hard to audit, whereas classical algorithms trained on tabular descriptors are cheap to train, easy to reproduce and, in several cases, directly interpretable. This work therefore concentrates on classical methods and asks how far they can go on a compact, well-known dataset.

Three concerns shape the design. First, errors are not equally costly: a malignant mass labelled benign delays treatment, whereas a benign mass labelled malignant normally triggers further testing. Recall-weighted measures have accordingly been advocated for breast tumour classification [6]; here the F2 score drives model selection and missed malignancies are counted explicitly. Second, small clinical datasets invite optimistic estimates

when preprocessing or tuning touches the test data, a widespread cause of irreproducible results [7]. Third, on a dataset of 569 samples a single train-test split leaves only about 40 malignant test cases, so one misclassification moves recall by more than two percentage points and the ranking of similar models can change with the split. The study therefore uses repeated nested cross-validation [8], which keeps tuning separate from testing and evaluates every model on the same 50 test folds, and judges differences with significance tests designed for cross-validated comparisons.

The contributions of this paper are as follows:

- From-scratch ID3 and C4.5 classifiers that follow the scikit-learn interface.
- A like-for-like comparison of eight classifiers under identical folds, tuning protocol and metrics, repeated ten times, with variance-corrected significance tests.
- An analysis of the error counts, hard cases and feature importance behind the models' decisions.

LITERATURE SURVEY

I. Classical classifiers on the Wisconsin data

Many studies benchmark conventional classifiers on the Wisconsin data. Kadhim and Kamil [9] compared eleven classifiers using specificity, sensitivity, precision, accuracy and F1, and found extremely randomised trees best, with an F1-score of 96.77%. Hossin *et al.* [10] evaluated logistic regression, random forest, KNN, decision tree, AdaBoost, SVM, gradient boosting and Gaussian naive Bayes with univariate and recursive feature selection; logistic regression and SVM reached the highest test accuracy (99.12%). Assiri *et al.* [6] ranked eight algorithms by the F3 score, which prioritises false negatives, and combined the three best through majority voting, reporting 99.42% accuracy on a 70:30 split. Osman and Aljahdali [11] applied an ensemble boosting scheme built on radial-basis-function networks to breast cancer screening.

II. Boosting and explainability

Gradient-boosted trees are frequently reported as strong learners on tabular data. Suresh *et al.* [12] trained an XGBoost classifier on the Wisconsin data (accuracy 98.42%) and used SHAP values to show that perimeter and concave-point features influenced its decisions most. Dutta *et al.* [13] likewise paired XGBoost with SHAP and LIME explanations, and Bentéjac *et al.* [14] compared the XGBoost, LightGBM and CatBoost implementations of gradient boosting. Rudin [15] argues that for high-stakes decisions inherently interpretable models should be preferred over post hoc explanations of black boxes, which motivates keeping transparent baselines in the comparison.

III. Decision trees, forests and margin classifiers

ID3 [16] grows trees on categorical attributes by choosing, at each node, the attribute with the highest information gain; Khedr *et al.* [17] proposed refinements to the trees it produces. C4.5 and CART, together with KNN and SVM, are among the most influential data-mining algorithms [18]: C4.5 extends ID3 with the gain ratio and thresholds for numeric attributes, whereas CART builds binary trees with the Gini index. Random forests [19] average many randomised trees to reduce the variance of a single tree, and Rokach [20] surveys two decades of decision-forest research. The support vector machine [21] separates classes with a maximum-margin boundary.

IV. Comparing classifiers fairly

Reported accuracies on the Wisconsin data typically come from one split, and differences of a fraction of a percentage point are often read as a ranking. Nadeau and Bengio [22] showed that the usual paired t-test over cross-validation folds is overconfident because the training sets overlap, and proposed a corrected variance; Bouckaert and Frank [23] found the corrected test on repeated cross-validation to be the most replicable choice. Demšar [24] recommends rank-based summaries, and Cawley and Talbot [25] show that tuning hyperparameters on the data used for evaluation biases the result, which nested cross-validation avoids.

V. Research gap

In the studies above, the original tree-induction algorithms ID3 and C4.5 are rarely examined explicitly, models are often tuned and scored under different splits, accuracy tends to be the headline figure, and differences are seldom tested for significance. The present study addresses these points with one protocol for all models, recall-oriented model selection, corrected significance tests and an analysis of the cases the models get wrong.

OBJECTIVES

The objectives of the project are the following:

1) Implement ID3 and C4.5 from scratch and compare them with CART, Random Forest and XGBoost, alongside Logistic Regression, KNN and SVM baselines.
2) Evaluate every model under identical conditions with accuracy, precision, recall, specificity, F1, ROC-AUC and confusion matrices, with particular attention to missed malignant cases.
3) Tune hyperparameters without touching the data used for evaluation, and test whether differences between models are statistically significant.
4) Analyse false negatives, false positives and the importance of individual diagnostic features.
5) Identify a model that balances predictive performance, stability and interpretability.

Data Source

The experiments use the Wisconsin Diagnostic Breast Cancer (WDBC) dataset created by Wolberg, Street and Mangasarian at the University of Wisconsin [3] and distributed through the UCI Machine Learning Repository. The working file, cleaned_data.csv, contains 569 samples (357 benign, 212 malignant) described by 30 real-valued features: the mean, standard error and worst (largest) value of ten nuclear characteristics, namely radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension. The target column *diagnosis* is coded 1 for malignant and 0 for benign. Checks found no missing values and no duplicate rows.

Exploratory analysis (Figure 1) shows moderate class imbalance (37.3% malignant) and clear class separation on size-related features such as perimeter_worst. Many features are strongly correlated: 21 feature pairs have an absolute correlation above 0.90, and radius, perimeter and area are almost interchangeable (r = 0.998 between radius_mean and perimeter_mean). Several error-term features, notably area_se, have heavy right tails. These properties matter for the scaling choices and for reading feature importance.

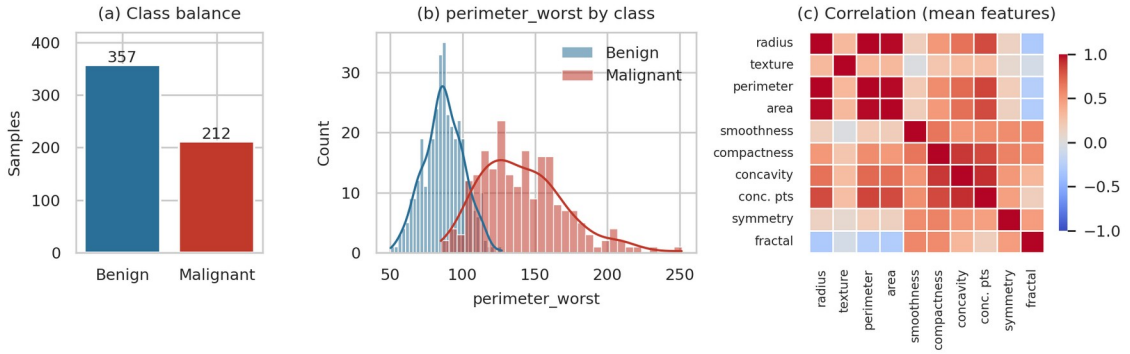


FIGURE 1. EXPLORATORY DATA ANALYSIS OF THE WDBC DATA. PANEL (A) SHOWS THE CLASS BALANCE, PANEL (B) THE DISTRIBUTION OF PERIMETER_WORST FOR EACH CLASS, AND PANEL (C) THE CORRELATION MATRIX OF THE TEN MEAN-VALUE FEATURES; THE NEAR-IDENTICAL RADIUS, PERIMETER AND AREA ROWS ILLUSTRATE THE STRONG COLLINEARITY.

Methodology

I. *Evaluation protocol*

The primary evaluation is a repeated nested cross-validation on all 569 samples [8], [25]. The outer loop is a stratified five-fold split repeated ten times with different shuffles (seed 42), which gives 50 outer folds of about 114 test and 455 training samples; across each repeat every sample is tested exactly once, so each sample is tested ten times by every model. All eight models receive exactly the same 50 outer folds, which makes the comparison paired. Inside each outer training part, hyperparameters are selected by an inner stratified five-fold grid search, so the outer test fold is never used for tuning. The selected configuration is refitted on the outer training part and scored once on the outer test fold.

As a secondary check that mirrors the usual practice, the data were also divided once into a stratified 80/20 training/test split (seed 42; 455 training and 114 test samples, 42 of them malignant), with tuning by five-fold cross-validation on the training part only.

II. *Preprocessing*

Logistic Regression, KNN and SVM are sensitive to feature scale, so a standard scaler (zero mean, unit variance) was placed inside a pipeline with the classifier; within every cross-validation loop it is refitted on the training data only, which prevents leakage [7]. Tree-based models are invariant to monotone rescaling and use the raw features.

III. *Classifiers*

ID3. Each feature is first discretised into quantile bins (the number of bins is a hyperparameter), because ID3 handles categorical attributes only. At every node the attribute with the highest information gain is chosen and a multiway split is made, one branch per bin. Writing p_i for the proportion of class i, the entropy of a sample set S and the information gain of splitting it on attribute A are

$$H(S) = -\sum_i p_i \log_2 p_i, \tag{1}$$

$$IG(S, A) = H(S) - \sum_v (|S_v| / |S|)\, H(S_v). \tag{2}$$

An attribute is used at most once on any root-to-leaf path, and growth stops at pure nodes, at a depth limit or when a node has too few samples.

C4.5. Numeric attributes are split with binary thresholds placed midway between consecutive sorted values. Information gain favours attributes with many outcomes, so C4.5 divides it by the split information to obtain the gain ratio:

$$SI(S, A) = -\sum_v (|S_v| / |S|) \log_2(|S_v| / |S|), \tag{3}$$

$$GR(S, A) = IG(S, A) / SI(S, A). \quad (4)$$

Following Quinlan's heuristic, only attributes whose best gain is at least the average gain over all candidates compete on gain ratio. Overfitting is controlled by depth and leaf-size limits rather than C4.5's pessimistic post-pruning, and no missing-value handling is needed because the data are complete.

CART. Binary splits are chosen with the Gini impurity

$$Gini(S) = 1 - \sum_i p_i^2, \quad (5)$$

and trees are pruned by cost-complexity pruning, with the pruning strength tuned. *Random Forest* [19] grows many randomised trees on bootstrap samples and averages their votes. *XGBoost* builds trees sequentially, each fitted to correct the errors of its predecessors under a log-loss objective [14]. *Logistic Regression* (L1 or L2 regularised), *KNN* [18] (distance-weighted or uniform voting) and *SVM* [21] (linear or RBF kernel) complete the set.

IV. Hyperparameter tuning

Every inner search uses the grids in Table 1 and stratified five-fold cross-validation [26]. The selection criterion is the F2 score, given by (9) with $\beta = 2$, which weights recall four times as heavily as precision. Because tuning is repeated in each of the 50 outer folds, the chosen settings can differ between folds; Table 1 reports the most frequent choice.

V. Evaluation metrics

With malignant as the positive class, and *TP*, *TN*, *FP* and *FN* denoting the confusion-matrix counts, the metrics are

$$\text{Precision} = TP / (TP + FP), \quad (6)$$

$$\text{Recall} = TP / (TP + FN), \quad (7)$$

$$\text{Specificity} = TN / (TN + FP), \quad (8)$$

$$F_\beta = (1 + \beta^2) \cdot P \cdot R / (\beta^2 \cdot P + R), \quad (9)$$

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{D}, \quad (10)$$

where P and R are precision and recall and $D = (TP + FP)(TP + FN)(TN + FP)(TN + FN)$. Accuracy, balanced accuracy (the mean of recall and specificity), F1 ($\beta = 1$), the area under the ROC curve [27] and the area under the precision-recall curve (PR-AUC) are also reported, following the measures catalogued in [28]. The Matthews correlation coefficient (MCC) uses all four cells of the confusion matrix and is more informative than accuracy or F1 on imbalanced data [29], so it serves, with F2, as a headline metric. Every metric is computed on each outer test fold; errors are additionally counted per pass, that is, summed over the five folds of one repeat, so that they refer to all 212 malignant and 357 benign cases.

VI. Statistical comparison

Cross-validation folds share training data, so their scores are correlated and the ordinary paired t-test understates the variance. Following Nadeau and Bengio [22], the difference d_j between two models on outer fold $j = 1, ..., J$ is tested with

$$t = \bar{d} / \sqrt{[(1/J + n_2/n_1) \cdot \sigma^2]}, \quad (11)$$

where $\bar{d}$ and σ^2 are the mean and variance of the differences, $J = 50$, $n_2/n_1 = 0.25$ is the ratio of test to training size, and t has $J - 1$ degrees of freedom; this corrected repeated cross-validation test has good replicability [23]. Each model is compared with the best one, and the seven p-values per metric are adjusted with the Holm procedure [30]. The same variance correction gives the 95% confidence intervals. Mean ranks over the 50 folds and a Friedman test are reported as a descriptive summary [24].

VII. Feature importance

Impurity-based importances were taken from ID3, C4.5, CART, Random Forest and XGBoost, and the absolute coefficients of the standardised Logistic Regression model were used for that model; importances are normalised to sum to one and are computed on the single-split models.

IMPLEMENTATION DETAILS

The code is written in Python 3.13 with scikit-learn 1.9.1 [31], XGBoost 3.4.1, NumPy, pandas, SciPy, Matplotlib and Seaborn. The script tree_models.py holds the from-scratch ID3Classifier and C45Classifier, which implement fit, predict, predict_proba and feature_importances_ and derive from the scikit-learn base classes, so that they work unchanged inside GridSearchCV. ID3 computes its quantile bin edges directly with NumPy, because the default binning of scikit-learn's KBinsDiscretizer changed in version 1.9, and C4.5 scores all candidate thresholds of a feature at once from cumulative class counts, which produces the same trees about 30 times faster.

The script nested_cv_comparison.py runs the repeated nested cross-validation: 50 outer folds, eight models and 245 hyperparameter configurations, about 61,000 model fits in all, which took 32 minutes on two CPU cores. It saves every fold's metrics and every out-of-fold prediction, and analyse_nested_cv.py computes the summaries, tests and figures. The single-split results come from compare_all_models.py, with the split written by make_split.py; they were identical under scikit-learn 1.8 and 1.9.1. All random seeds are fixed at 42.

TABLE 1. HYPERPARAMETER GRIDS AND THE SETTING CHOSEN MOST OFTEN IN THE 50 OUTER FOLDS

Model	Grid searched	Most frequent choice (folds of 50)
Logistic Regression	C ∈ {0.001, 0.01, 0.1, 1, 10, 100}; penalty ∈ {L1, L2}; class weight ∈ {none, balanced}	C = 0.1, L2, balanced (23)
KNN	k ∈ {1, 3, 5, 7, 9, 11, 15, 21}; weights ∈ {uniform, distance}; metric ∈ {Euclidean, Manhattan}	k = 3, Manhattan, uniform (15)

Model	Grid searched	Most frequent choice (folds of 50)
SVM	linear: C ∈ {0.01, 0.1, 1, 10, 100}; RBF: C ∈ {0.1, 1, 10, 100}, γ ∈ {scale, 0.001, 0.01, 0.1}	RBF, C = 10, γ = scale (11)
ID3	bins ∈ {3, 5, 8}; max depth ∈ {3, 5, 8, 12}; min split ∈ {2, 10}	3 bins, depth 3, min split 2 (13)
C4.5	max depth ∈ {3, 5, 8, 12}; min split ∈ {2, 10}; min leaf ∈ {1, 3, 5}	depth 5, min split 2, min leaf 5 (14)
CART	max depth ∈ {3, 5, 8, none}; min split ∈ {2, 10}; min leaf ∈ {1, 3, 5}; ccp_alpha ∈ {0, 0.005, 0.01}	depth 8, min leaf 1, min split 2, ccp_alpha 0 (8)
Random Forest	trees ∈ {100, 300}; max depth ∈ {none, 5, 10}; min leaf ∈ {1, 3}; max features ∈ {sqrt, log2}	100 trees, depth none, min leaf 1, log2 (9)
XGBoost	trees ∈ {100, 300}; max depth ∈ {2, 3, 5}; learning rate ∈ {0.05, 0.1}; subsample ∈ {0.8, 1.0}	300 trees, depth 2, rate 0.1, subsample 0.8 (11)

Results and Discussion

I. *Repeated nested cross-validation*

Table 2 summarises the 50 outer folds and Figure 2 shows the means of recall, F2 and MCC with their corrected 95% confidence intervals. Logistic Regression achieved the highest mean accuracy (0.974), recall (0.958), F2 (0.961) and MCC (0.945), with specificity 0.983 and ROC-AUC 0.994. SVM followed very closely (recall 0.954, MCC 0.941, ROC-AUC 0.994), then XGBoost (recall 0.945, MCC 0.931). Random Forest and KNN formed a middle group with recall near 0.93, and the three single decision trees came last, with recall 0.902 (ID3), 0.901 (C4.5) and 0.897 (CART) and MCC around 0.85. The ordering is the same for nearly every metric, and the trees also have the widest spread across folds.

TABLE 2. REPEATED NESTED CROSS-VALIDATION (5 FOLDS × 10 REPEATS, IDENTICAL FOLDS FOR ALL MODELS): MEAN WITH STANDARD DEVIATION OVER THE 50 OUTER FOLDS IN BRACKETS; FN AND FP ARE ERRORS PER PASS OVER ALL 569 SAMPLES (212 MALIGNANT, 357 BENIGN); BEST VALUE PER COLUMN IN BOLD

Model	Accuracy	Balanced accuracy	Precision	Recall	Specificity	F1	F2	MCC	ROC-AUC	PR-AUC	FN per pass	FP per pass
Logistic Regression	**0.974 (0.014)**	**0.971 (0.016)**	**0.973 (0.026)**	**0.958 (0.029)**	**0.983 (0.016)**	**0.965 (0.019)**	**0.961 (0.024)**	**0.945 (0.030)**	**0.994 (0.007)**	**0.993 (0.007)**	**8.9**	**5.9**
KNN	0.962 (0.017)	0.955 (0.020)	0.971 (0.031)	0.928 (0.038)	**0.983 (0.020)**	0.948 (0.024)	0.936 (0.031)	0.920 (0.036)	0.974 (0.021)	0.957 (0.041)	15.3	6.2
SVM	0.972 (0.014)	0.968 (0.016)	0.971 (0.026)	0.954 (0.031)	**0.983 (0.016)**	0.962 (0.020)	0.957 (0.025)	0.941 (0.030)	**0.994 (0.006)**	**0.993 (0.006)**	9.7	6.2
ID3	0.933 (0.018)	0.927 (0.022)	0.917 (0.034)	0.902 (0.043)	0.951 (0.022)	0.909 (0.026)	0.905 (0.035)	0.857 (0.039)	0.944 (0.027)	0.897 (0.050)	20.7	17.6
C4.5	0.931 (0.022)	0.925 (0.025)	0.916 (0.041)	0.901 (0.048)	0.950 (0.028)	0.907 (0.030)	0.903 (0.038)	0.854 (0.046)	0.939 (0.028)	0.891 (0.048)	21.0	18.0
CART	0.929 (0.022)	0.923 (0.025)	0.915 (0.043)	0.897 (0.049)	0.949 (0.028)	0.904 (0.031)	0.899 (0.040)	0.850 (0.047)	0.932 (0.029)	0.885 (0.048)	21.9	18.3
Random Forest	0.957 (0.019)	0.952 (0.023)	0.955 (0.033)	0.930 (0.048)	0.973 (0.021)	0.941 (0.026)	0.934 (0.038)	0.909 (0.040)	0.990 (0.008)	0.988 (0.008)	14.8	9.6
XGBoost	0.968 (0.014)	0.963 (0.018)	0.968 (0.023)	0.945 (0.035)	0.981 (0.014)	0.956 (0.020)	0.949 (0.028)	0.931 (0.031)	0.993 (0.007)	0.992 (0.007)	11.7	6.7

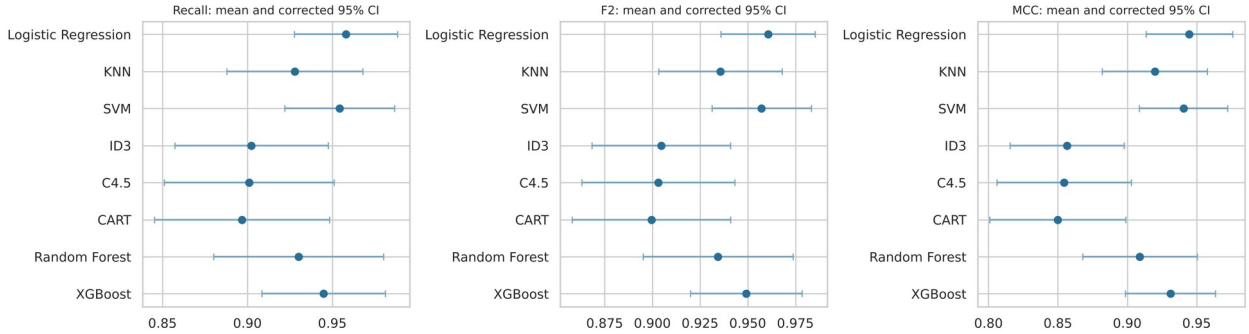


FIGURE 2. MEAN RECALL, F2 AND MCC OVER THE 50 OUTER FOLDS WITH NADEAU-BENGIO CORRECTED 95% CONFIDENCE INTERVALS. LOGISTIC REGRESSION, SVM AND XGBOOST OVERLAP ALMOST COMPLETELY, WHILE THE INTERVALS OF THE THREE SINGLE DECISION TREES LIE CLEARLY LOWER.

II. *Statistical significance*

Table 3 compares every model with Logistic Regression using the corrected t-test of (11). None of the differences to SVM, XGBoost, Random Forest or KNN is significant on any metric (all Holm-adjusted $p \geq 0.30$), so the data do not support a strict ranking among these five. ID3, C4.5 and CART, in contrast, are significantly worse on MCC ($p = 0.014$, 0.001 and 0.005), accuracy and ROC-AUC. On

recall alone, no difference survives the correction, because recall varies strongly between folds that contain only about 42 malignant cases. The correction matters: an uncorrected paired t-test would declare Logistic Regression significantly better than XGBoost on MCC ($p = 0.018$), a conclusion the corrected test does not support. Mean ranks over the folds lead to the same picture; by MCC they are 2.35 for Logistic Regression, 2.62 for SVM, 2.99 for XGBoost, 3.63 for KNN, 4.17 for Random Forest and 6.66–6.79 for the three trees (Friedman test, $p < 0.001$).

TABLE 3. DIFFERENCE TO LOGISTIC REGRESSION (LOGISTIC REGRESSION MINUS MODEL) WITH HOLM-ADJUSTED P-VALUE OF THE CORRECTED T-TEST IN BRACKETS; * MARKS $p < 0.05$

Model	MCC	F2	ROC-AUC
KNN	+0.025 (0.829)	+0.025 (0.428)	+0.020 (0.308)
SVM	+0.004 (1.000)	+0.003 (0.903)	-0.000 (1.000)
ID3	+0.088 (0.014)*	+0.056 (0.083)	+0.050 (0.004)*
C4.5	+0.090 (0.001)*	+0.057 (0.017)*	+0.055 (0.002)*
CART	+0.095 (0.005)*	+0.061 (0.040)*	+0.062 (0.002)*
Random Forest	+0.036 (0.498)	+0.026 (0.519)	+0.004 (1.000)
XGBoost	+0.014 (1.000)	+0.012 (0.903)	+0.001 (1.000)

III. Missed malignancies and false alarms

The last two columns of Table 2 translate the metrics into patients. Each pass classifies every sample once, and Logistic Regression missed 8.9 of the 212 malignant cases per pass while raising 5.9 false alarms among the 357 benign cases; SVM missed 9.7 and XGBoost 11.7. KNN and Random Forest missed about 15, and the single trees about 21 malignant cases per pass, more than twice as many as Logistic Regression, while also raising about three times as many false alarms (17.6–18.3). For a screening aid the difference between 9 and 21 missed cancers in 212 is the most tangible result of the study.

IV. Agreement with a single held-out split

Table 4 reports the secondary 80/20 split. Logistic Regression was again best, with accuracy 0.991, recall 0.976 and a single missed malignancy, followed by SVM and XGBoost (three misses each); C4.5 and CART missed five. The ordering matches the nested results, but the single-split figures for the best models are higher than their cross-validated means (recall 0.976 against 0.958 for Logistic Regression), which illustrates how much one split of 42 malignant cases can flatter a model. The nested estimates are therefore the ones to quote.

TABLE 4. SINGLE HELD-OUT SPLIT (114 TEST SAMPLES: 42 MALIGNANT, 72 BENIGN)

Model	Acc.	Prec.	Recall	Spec.	ROC-AUC	FN	FP
Logistic Regression	0.991	1.000	0.976	1.000	0.998	1	0
KNN	0.965	1.000	0.905	1.000	0.974	4	0
SVM	0.974	1.000	0.929	1.000	0.993	3	0
ID3	0.956	0.951	0.929	0.972	0.972	3	2
C4.5	0.921	0.902	0.881	0.944	0.920	5	4
CART	0.921	0.902	0.881	0.944	0.943	5	4
Random Forest	0.965	1.000	0.905	1.000	0.996	4	0
XGBoost	0.974	1.000	0.929	1.000	0.993	3	0

V. Error analysis

Because every sample was tested ten times by each of the eight models, the out-of-fold predictions show which cases are intrinsically hard. Of the 569 samples, 380 were never misclassified by any model in any repeat, and only 16 (12 malignant, 4 benign) were misclassified in at least half of the 80 model-repeat predictions. Three malignant samples (zero-based rows 40, 135 and 297 of cleaned_data.csv) were missed by every model in every repeat: their radius_mean values of 13.4, 12.8 and 11.8 lie among the smallest 10% of malignant masses, well below the malignant median of 17.3 and close to the benign median of 12.2, and their concave points_mean values (0.020–0.035) are close to the benign median of 0.023 and far below the malignant median of 0.086. The single malignant case that Logistic Regression missed on the held-out split (row 73 of the data file, radius_mean 13.8, area_mean 584.1) belongs to the same group and was misclassified in 96% of all predictions. These small, smooth malignant masses form a region of the feature space where the two classes overlap, which bounds what any classifier on these 30 features can achieve.

VI. Threshold analysis

Recall can be traded against precision by moving the decision threshold. For Logistic Regression on the single split, out-of-fold training predictions showed that lowering the threshold from 0.5 to 0.2 raised cross-validated recall from 0.959 to 0.994 while precision fell from 0.976 to 0.816; on the test set this removed the one false negative at the cost of seven false positives. Such a trade is a clinical decision, and the threshold must be chosen on training-side predictions, never on the test set.

VII. Feature importance

Figure 3 shows that the single trees rely overwhelmingly on one attribute: perimeter_worst carries 78.6% of CART's importance, 70.7% of ID3's and 62.3% of C4.5's. Random Forest spreads importance over concave points_worst (0.142), area_worst (0.119), perimeter_mean (0.114) and radius_worst (0.109), and XGBoost over radius_worst, perimeter_worst and concave points_mean (about 0.22 each). In Logistic Regression the largest standardised coefficients belong to texture_worst (0.62), radius_worst (0.51) and radius_se (0.49). Because radius, perimeter and area are almost perfectly correlated, the ranking among them is arbitrary and should be read as a group. Overall, size and concavity descriptors of the

"worst" nuclei dominate, which agrees with the SHAP analyses of XGBoost in [12] and explains why small malignant masses with low concavity are the ones that are missed.

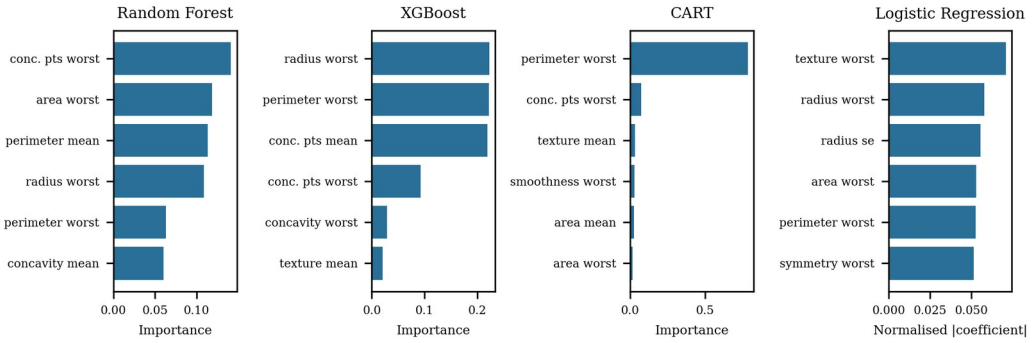


FIGURE 3. TOP SIX FEATURES FOR FOUR MODELS TRAINED ON THE SINGLE SPLIT. BARS SHOW NORMALISED IMPURITY IMPORTANCE FOR RANDOM FOREST, XGBOOST AND CART, AND THE ABSOLUTE STANDARDISED COEFFICIENT FOR LOGISTIC REGRESSION; SIZE AND CONCAVITY DESCRIPTORS OF THE WORST NUCLEI DOMINATE IN EVERY MODEL.

VIII. Model choice and interpretability

Logistic Regression, SVM and XGBoost are statistically indistinguishable on this dataset, so the choice among them rests on other grounds. Logistic Regression is the most attractive: it has the best mean on every headline metric, the most stable hyperparameters (the same setting was chosen in 23 of 50 folds, against 11 for SVM and XGBoost, whose 50 folds produced 11 and 20 distinct settings), and coefficients that can be inspected directly, in line with the argument for interpretable models in [15]. The single trees are the easiest to read as rules, yet they were significantly less accurate and the least stable, missing more than twice as many malignant cases.

IX. Limitations

The study uses one public dataset whose features were extracted in advance and no external clinical validation; repeated cross-validation measures how well the models generalise within this population, not to other hospitals or imaging set-ups. The significance tests reduce, but cannot remove, the dependence between overlapping folds. The models are an academic comparison and must not be used for medical decision-making.

CONCLUSION AND FUTURE SCOPE

This study compared ID3, C4.5, CART, Random Forest, XGBoost, Logistic Regression, KNN and SVM on the WDBC dataset under one leakage-free protocol: ten-times repeated nested cross-validation with identical folds for every model, recall-weighted tuning and variance-corrected significance tests. Logistic Regression achieved the best mean performance (accuracy 0.974, recall 0.958, MCC 0.945) and missed about 9 of 212 malignant cases per pass, but SVM and XGBoost were statistically indistinguishable from it. The three single decision trees, including the from-scratch ID3 and C4.5, were significantly worse and missed about 21. Most remaining errors come from a small group of small, low-concavity malignant masses that overlap with benign tissue in feature space. A single train-test split reproduced the ordering but overstated the best model's recall, which shows why repeated evaluation and significance testing are needed before ranking similar classifiers.

Future work includes validating the models on independent cohorts; feature selection or principal component analysis to handle collinearity; threshold selection and probability calibration for a stated false-negative budget inside the nested loop; SHAP-based explanations; cost-sensitive learning; extending the from-scratch C4.5 with pessimistic pruning and missing-value handling; and the optional C++ implementation of Logistic Regression with gradient descent proposed in the project synopsis.

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About the Authors

Udit Asthana is an undergraduate student pursuing a B.Tech in Computer Science and Engineering at Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal (Roll No. 31, Reg. No. 240905310).

C. Nagesh Prabhu is an undergraduate student pursuing a B.Tech in Computer Science and Engineering at Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal (Roll No. 2, Reg. No. 240905014).

Mahad Sheikh is an undergraduate student pursuing a B.Tech in Computer Science and Engineering at Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal (Roll No. 36, Reg. No. 240905382).